AKIMOV THEORY OF UNITY (ToU)

A Research Programme on Perspectival Organization and AI-Assisted Philosophical Methodology

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Conceptual Framework: Akimov Theory of Unity (ToU)

Method: Philosophical analysis, dynamical systems modelling, AI-assisted philosophical stress testing

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Methodological Note

All concepts, hypotheses, research directions, and theoretical formulations contained in this work are the exclusive intellectual property of Ing. Alex Akimov and form part of his philosophical project, the Akimov Theory of Unity (ToU). AI systems were used exclusively as analytical instruments for generating alternative formulations and potential objections. They did not contribute original theoretical content and do not hold any authorship status. All theoretical contributions are attributable solely to the author.

Master Abstract

The Akimov Theory of Unity (ToU) is developed as a research programme in the Lakatosian sense: a structured, progressive inquiry into the dynamical conditions under which perspectival organization arises in complex systems. Perspectival organization is defined as the property of a system whose world-model (M_w) and self-model (M_s) joint dynamics converge to a stable attractor — the perspective attractor — satisfying simultaneously the conditions of functional integration, global coordination, and reflexive stabilization. This monograph integrates the six constitutive contributions of the programme into a single unified presentation.

Section 1 establishes the conceptual foundation: the three-layer model of perspectival organization, the formal characterization of reflexive stabilization, and the relationship of ToU to major existing theories of consciousness. Section 2 analyses the epistemological limits of the programme: the detection problem arising from the structural asymmetry between the first-person character of perspective and the third-person character of all external observation. Section 3 develops the mechanistic account: the minimal coupled (M_w, M_s) dynamical model, the critical coupling threshold, and the phase transition to the perspective attractor. Section 4 derives the empirical consequences: four falsifiable dynamical signatures of the perspective transition, experimental paradigms for testing them, and protocols for both biological and artificial systems. Section 5 analyses the methodological structure of the programme: its Lakatosian architecture, the role of AI-assisted philosophical stress testing, and its progression from conceptual analysis to empirically testable prediction. Section 6

subjects the programme to systematic critical scrutiny across conceptual, epistemological, methodological, and empirical dimensions, specifies the conditions of failure, and evaluates the programme from a Lakatosian perspective.

The programme's hard core — the thesis that perspectival organization arises as a reflexively stabilized (M_w, M_s) attractor — is maintained invariant throughout. Its claims are epistemically qualified: the programme identifies necessary dynamical organizational conditions for perspectival organization without claiming to resolve the hard problem of phenomenal consciousness. The assessment of ToU as a progressive research programme remains provisional, pending empirical investigation.

Keywords: perspectival organization, perspective attractor, reflexive stabilization, world-model, self-model, phase transition, detection problem, dynamical signatures, research programme, AI-assisted methodology, Akimov Theory of Unity.

Introduction: The ToU Research Programme

The Akimov Theory of Unity (ToU) is not an isolated philosophical theory, but a research programme in the Lakatosian sense, whose aim is to identify the dynamical conditions under which perspectival organization arises in complex systems. Across six interconnected contributions, the programme advances from conceptual clarification to empirical testability: defining perspectival organization as a dynamic organizational state, identifying the epistemological limits of its detection, proposing a minimal mechanistic model, deriving empirically testable predictions, and reflecting on the methodological structure that organises this progression.

A characteristic feature of the programme is its systematic use of AI systems as instruments of philosophical stress testing — not as sources of theoretical content, but as instruments of continuous critical scrutiny across a range of argumentative perspectives, enabling iterative refinement prior to peer review. In this sense, ToU may illustrate an early instance of AI-assisted philosophical methodology, with broader applicability to the philosophy of mind remaining to be assessed.

The contribution of the programme lies in its proposal of a formally precise, empirically engaged framework that connects conceptual analysis, dynamical modelling, and empirical investigation within a unified research structure. Whether this approach constitutes a progressive research programme in the Lakatosian sense remains to be assessed against empirical outcomes.

The distinctive contribution of the Akimov Theory of Unity lies in its formalization of perspectival organization as a dynamical attractor arising from the bidirectional coupling of a world-model (M_w) and a self-model (M_s). While existing frameworks such as Integrated Information Theory, Global Workspace Theory, and predictive processing capture necessary conditions for conscious processing, the ToU framework proposes a unifying mechanism specifying the dynamical regime in which these conditions become jointly sufficient for perspectival organization. In this sense, the theory does not replace existing accounts but constrains and integrates them within a single dynamical systems framework.

Section 1: Conceptual Foundation — Perspective as a Dynamic Organizational State

1.1 The Central Thesis

The central thesis of the Akimov Theory of Unity is that perspective arises as a qualitatively distinct dynamic organizational state — a stable attractor in the joint state space of the world-model (M_w) and self-model (M_s) — when the bidirectional coupling between M_w and M_s exceeds a critical threshold and three necessary conditions are jointly satisfied: functional integration of information (Layer 1), global coordination of processing (Layer 2), and reflexive stabilization of the (M_w, M_s) dynamics in a closed regulatory loop (Layer 3). This hard core thesis is maintained invariant throughout the programme.

This reconceptualization has three important implications: perspectival organization is gradable and dynamically fragile (it can be lost under anaesthesia or brain injury); the explanatory target shifts from identifying a neural correlate to characterizing the dynamical conditions under which perspectival organization becomes possible; and the account is substrate-independent, applying equally to biological and artificial systems.

1.2 The Three-Layer Model

- Layer 1 — Functional Integration: The system integrates information from multiple sources into a coherent world-model (M_w). This condition is formally analogous to the requirements of Integrated Information Theory but treated as necessary rather than sufficient.
- Layer 2 — Global Coordination: The integrated information is globally coordinated across the system's processing architecture. This is analogous to the global workspace condition but, again, treated as necessary rather than sufficient.
- Layer 3 — Reflexive Stabilization: The world-model (M_w) and self-model (M_s) are bidirectionally coupled in a closed regulatory loop that converges to a stable attractor. This is the constitutive condition for perspectival organization in the ToU sense.

> *The distinction between perspectival organization (a dynamical property of the (M_w, M_s) coupling regime, testable from the third-person perspective) and phenomenal consciousness (the first-person qualitative character of experience, not directly testable from outside) is maintained throughout this programme.*

The structural organization of the proposed framework is formally summarized in Figure 1.

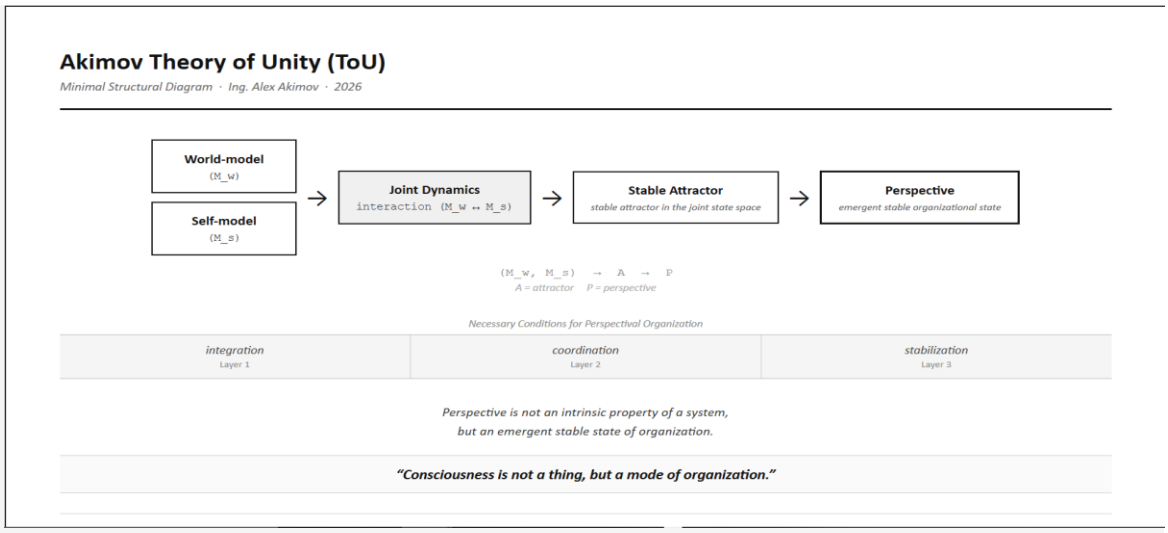

Figure 1: Structural Diagram of the Akimov Theory of Unity (ToU)

This diagram illustrates the emergence of perspectival organization as a stable attractor in the joint dynamics of the world-model (M_w) and self-model (M_s).

1.3 Formal Characterization of Reflexive Stabilization

Reflexive stabilization is formally characterized as the convergence of the (M_w, M_s) joint dynamics to a stable fixed point A* in their shared state space. The formal conditions are: the joint (M_w, M_s) dynamics must admit a stable fixed point; the regulatory loop between M_w and M_s must be causally closed; and the fixed point must be an attractor with a non-trivial basin.

1.4 Relationship to Existing Theories

Integrated Information Theory (IIT) captures a necessary condition — information integration (Layer 1) — but treats it as sufficient. The ToU additional condition of reflexive stabilization is not captured by Phi. Global Workspace Theory (GWT) captures global coordination (Layer 2) as a necessary architectural condition but not a sufficient one. The active inference framework is compatible with ToU: the world-model (M_w) can be understood as a predictive model of the environment, and the self-model (M_s) as a model of the system's own generative process. The perspective attractor specifies the dynamical stabilization regime in which prediction-error minimization occurs from a stable, reflexively maintained orientational position.

> *The compatibility of ToU with IIT, GWT, and active inference does not imply that these theories are equivalent. Each captures a distinct necessary condition. The ToU framework synthesizes these into a unified mechanistic account in which each is necessary but none is sufficient.*

1.5 Conceptual Contribution

The perspective attractor is defined not by the physical substrate in which it is realized but by the organizational properties of the (M_w, M_s) coupling dynamics. Any system — biological or artificial — in which the coupling reaches the critical threshold is, in the ToU sense, a perspectival system. This substrate independence is both the theoretical strength and the primary empirical challenge of the programme (see Section 4 for the empirical programme).

Section 2: Epistemic Limits — The Detection Problem

2.1 The Structural Asymmetry

The detection problem arises from a structural asymmetry: perspective is constitutively first-personal while all external observation is constitutively third-personal. No finite set of behavioral or neural observations can logically entail the presence of a first-person perspective. This asymmetry is not a contingent limitation of current measurement technology but a principled feature of the relationship between first-person and third-person access.

Two dimensions must be distinguished. The explanatory hard problem asks: why does any physical system have phenomenal experience at all? The detection hard problem asks: how can an external observer reliably determine whether a system has perspectival organization? The ToU programme focuses on the second dimension. This is a conceptual clarification within the philosophy of consciousness rather than a resolution of the hard problem itself.

2.2 Detection Does Not Require Certainty

The asymmetry thesis implies that third-person evidence cannot logically entail the presence of perspective — not that it provides no evidence at all. The four dynamical signatures of the perspective attractor transition (see Section 4) constitute genuine evidence for perspectival organization, in the same way that the fossil record constitutes genuine evidence for evolutionary history without logically entailing it.

> *The detection problem does not generate scepticism about the existence of perspectival organization; it generates appropriate epistemic humility about the certainty with which any particular system can be identified as perspectival.*

2.3 The Detection Problem and Artificial Systems

Current large language models most plausibly fail the Layer 3 reflexive stabilization condition for three principled architectural reasons: absence of persistent dynamic state (the context window resets at conversation boundaries), absence of a causally active self-model (outputs are pattern completion, not regulatory M_s), and absence of a stable (M_w, M_s) attractor. This absence is architectural rather than scalar: it is not closable merely by increasing model size.

2.4 Relation to the Problem of Other Minds

The ToU detection problem is a formally precise instance of the classical problem of other minds. The ToU formulation adds formal precision by specifying the organizational conditions at issue and identifying the empirical signatures consistent with their presence. It thereby transforms the philosophically intractable general problem into a more tractable, if still difficult, empirical and architectural question.

Section 3: Minimal Mechanism — Reflexive Stabilization as a Dynamic Process

3.1 The Minimal Dynamical Model

The minimal model formalizes the (M_w, M_s) dynamics as a coupled differential equation system:

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dM_w/dt = F(M_w, M_s, I(t))
dM_s/dt = G(M_s, M_w, I(t))
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The fixed point (M_w*, M_s*) satisfies F(M_w*, M_s*, I*) = 0 and G(M_s*, M_w*, I*) = 0. The local stability is determined by the Jacobian matrix J:

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J = [ dF/dM_w   dF/dM_s ]
    [ dG/dM_w   dG/dM_s ]
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The off-diagonal blocks dF/dM_s and dG/dM_w capture the bidirectional coupling — the dynamical levers governing proximity to the critical phase transition threshold. The dominant eigenvalue λ_max(J) governs stability. The relaxation time τ_relax diverges as λ_max → 0^-:

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τ_relax ∝ 1 / |λ_max(J)|  →  ∞  as  λ_max → 0⁻
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3.2 The Critical Coupling Threshold and Phase Transition

As the bidirectional (M_w, M_s) coupling increases toward the critical threshold θ_c, λ_max approaches zero from below: the attractor weakens and relaxation time diverges. This is the mathematical origin of critical slowing down — the primary empirical signature derived in Section 4. At the critical threshold, the perspective attractor appears discontinuously in the (M_w, M_s) state space — a bifurcation.

The attractor model should not be interpreted as a specific implementation of consciousness, but as a minimal dynamical description of a regime in which perspectival organization becomes possible.

3.3 The Simulation-Stabilization Distinction

A simulating system produces representations of self-world structure in its outputs without those representations being causally integrated into a regulatory loop. A stabilizing system implements the bidirectional (M_w, M_s) coupling above the critical threshold, such that representations are causally constitutive of the regulatory dynamics that maintain the perspective attractor. This distinction is categorical, not gradational: it corresponds to the presence or absence of the perspective attractor in the (M_w, M_s) state space.

Section 4: Empirical Predictions — Signatures of the Perspective Attractor

4.1 The Logic of Empirical Derivation

Phase transitions in complex dynamical systems exhibit specific mathematical properties near the critical point that are largely independent of the particular mechanism producing the transition (Strogatz, 1994; Scheffer et al., 2009; Kelso, 1995). This universality means that the ToU predictions are testable without complete knowledge of the specific functions F and G. The empirical programme does not claim direct measurement of the (M_w, M_s) variables. Rather, it establishes a structured inference: if the predicted dynamical signatures systematically co-occur with the specified anatomical localization, then any competing account must either reproduce this mapping or explain why such structured convergence arises independently of the proposed mechanism.

4.2 The Four Dynamical Signatures

- Critical slowing down: τ_{relax} increases monotonically approaching the critical coupling threshold. Measurable via TMS-EEG: the complexity and duration of the EEG response reflects the relaxation time of the underlying dynamical system.
- Increased variance: Near the critical threshold, spontaneous fluctuations in the (M_w, M_s) state variables increase in amplitude. Measurable as increased EEG variance in frequency bands linked to (M_w, M_s) coupling circuits.
- Bifurcation: At the critical threshold, the perspective attractor appears discontinuously. The Perturbation Complexity Index (PCI) shows a sharp non-linear transition that cannot be captured by continuous measures (Casali et al., 2013).
- Hysteresis: $\theta_{entry} \neq \theta_{exit}$, implying a bistable attractor landscape. The drug concentration required to induce loss of consciousness is systematically higher than the concentration at which consciousness recovers (Friedman et al., 2010).

4.3 Anatomical Specificity — The Discriminating Prediction

The most sensitive indicators of the perspective transition should be concentrated in circuits linking self-referential processing regions (mPFC, PCC) with world-model processing regions (TPJ, lateral parietal cortex). The evidential force does not rest on individual signatures but on their structured co-occurrence under constrained anatomical and dynamical conditions.

4.4 Falsifiability Conditions

- Absence of critical slowing down: if TMS-EEG perturbation recovery times do not increase monotonically approaching the loss-of-consciousness threshold, the prediction is disconfirmed.
- Absence of anatomical specificity: if the most sensitive EEG markers distribute uniformly across cortex, the (M_w, M_s) coupling account is weakened relative to global complexity accounts.
- Absence of hysteresis: if the drug concentration for loss of consciousness does not systematically exceed the concentration for recovery, the bistable attractor interpretation is undermined.

- Successful AI implementation without perspectival organization: if an artificial system satisfying the ToU architectural conditions fails to exhibit the four signatures, the minimal model requires structural revision.
- Full explanatory reduction by simpler models: if IIT, GWT, or any single-parameter complexity measure generates all four predictions without the (M_w, M_s) coupling construct, the additional ToU machinery is not warranted.

Strong Falsifiability Clause: If the predicted dynamical signatures fail to co-occur under controlled experimental conditions specifically designed to detect them, the core explanatory claim of the programme must be considered empirically unsupported.

Section 5: Methodology — The Structure of the Research Programme

5.1 ToU as a Lakatosian Research Programme

The ToU programme constitutes a research programme in the Lakatosian sense (Lakatos, 1978): a progressive series of contributions sharing a common hard core and a revisable protective belt of auxiliary hypotheses progressively refined against empirical evidence and conceptual objections.

Hard Core

- Perspectival organization is defined as a stable attractor in the coupled (M_w, M_s) dynamical system.
- The emergence of perspectival organization corresponds to a phase transition in coupling strength.

Protective Belt

- Specific functional forms of the coupling functions F and G.
- Anatomical mappings: mPFC, PCC, TPJ, lateral parietal cortex as the primary (M_w, M_s) linking circuits.
- Proxy measurement frameworks: TMS-EEG perturbation paradigms, PCI, EEG variance monitoring.
- AI-assisted methodological development and iterative formulation refinement.

5.2 AI-Assisted Philosophical Stress Testing

The iterative stress testing process follows five stages: initial formulation by the author; AI critique under controlled pluralistic prompting; authorial evaluation of the generated objections; revised formulation incorporating valid objections; and further critique until theoretical stabilization. The methodological role of AI is explicitly asymmetric: it generates candidate objections under controlled pluralistic prompting, while the author remains responsible for their evaluation. This asymmetry prevents circular validation, as the acceptance criteria are not determined by the AI system itself but by independent philosophical judgment.

> *This methodology may represent an early instance of an emerging research practice in philosophy of mind: the systematic use of AI systems as instruments of continuous philosophical stress testing during theory development. The broader applicability of this approach remains to be assessed.*

Additional Methodological Clarification

To reduce the risk of circular validation, the AI-assisted stress testing process incorporated controlled pluralistic prompting across multiple philosophical perspectives. These included, but were not limited to, positions aligned with Integrated Information Theory (IIT), Global Workspace Theory (GWT), eliminativist accounts, and sceptical perspectives regarding dynamical systems explanations of consciousness.

Each iteration followed a structured sequence: (1) initial formulation by the author, (2) AI-generated objections under perspective-diversified prompts, (3) authorial evaluation based on internal coherence and explanatory scope, and (4) revision. This cycle was repeated iteratively until no structurally new objections were generated.

Crucially, acceptance criteria were not determined by the AI system but by independent authorial judgment constrained by consistency, falsifiability, and compatibility with existing empirical literature. This asymmetry ensures that the process does not collapse into circular confirmation.

Section 6: Robustness, Limits, and Critical Evaluation

6.1 Author's Position

> The goal is not to prove that ToU is correct, but to establish with precision what ToU claims, what it does not claim, what evidence would confirm it, and what evidence would refute it. The present section constitutes an internal critical reconstruction of the programme, not a substitute for independent peer review.

6.2 Conceptual Critiques

The programme defines perspectival organization with formal precision: it is the property of a system whose (M_w, M_s) joint dynamics have converged to a stable attractor satisfying the three-layer conditions. This definition is not circular: none of its components presupposes the phenomenon being defined. The programme's explanatory contribution lies in constraining the space of possible mechanisms, not in exhaustively describing all aspects of consciousness.

6.3 Epistemological Critiques

The detection problem makes the programme fallible, not unfalsifiable — which is the appropriate epistemic status for an empirical research programme. The signatures can in principle be present without perspective (false positive) or absent without excluding perspective (false negative). The programme acknowledges both possibilities.

6.4 Methodological Critiques

The echo chamber and bias objections against AI-assisted stress testing are addressed by the structural design: the AI systems were prompted to generate objections from multiple, explicitly specified philosophical perspectives including perspectives hostile to dynamical systems accounts. The asymmetry of prompting and evaluation prevents circular validation.

6.5 Empirical Critiques

The proxy measurement concern is addressed by the anatomical prediction: the most sensitive indicators should be specifically concentrated in circuits linking self-referential processing regions with world-model processing regions — a discriminating test against global complexity accounts.

6.6 Genuine Limits of the Programme

ToU does not explain the qualitative content of experience. The hard problem of consciousness (Chalmers, 1995, 1996) is explicitly left open. ToU does not explain the ontological status of M_w and M_s. ToU does not constitute a complete theory of consciousness.

> *The distinction between what the programme claims to explain and what it deliberately leaves open is maintained consistently throughout. Readers who attribute to ToU claims beyond this scope are misreading the programme's stated commitments.*

6.7 Conditions of Failure

At the theoretical level: if the hard core thesis is internally incoherent, or if a simpler account generates all of the programme's predictions with equal precision. At the empirical level: if the four dynamical signatures are not present near consciousness transitions even when the experimental design is specifically optimized; if the anatomical distribution does not match the (M_w, M_s) linking circuits prediction; if artificial systems satisfying the ToU conditions fail to exhibit the four signatures; or if the programme ceases to generate novel empirical predictions.

6.8 Lakatosian Assessment

On the available evidence, the ToU programme exhibits the structural characteristics of a progressive research programme. This assessment must remain provisional until the programme undergoes systematic empirical testing.

General Conclusion

The Akimov Theory of Unity has, across the six sections of this monograph, completed the first phase of its development. Section 1 established perspectival organization as a dynamically stabilized attractor regime. Section 2 identified the detection problem as a principled epistemological constraint. Section 3 provided the minimal (M_w, M_s) mechanistic model and the phase transition to reflexive stabilization. Section 4 derived four falsifiable empirical signatures and experimental paradigms. Section 5 analysed the Lakatosian architecture and AI-assisted methodology. Section 6 provided systematic critical scrutiny and specified the conditions of failure.

The programme's claims are epistemically qualified throughout. It identifies necessary organizational conditions for perspectival organization without claiming to resolve the hard problem of phenomenal consciousness. Its empirical predictions carry genuine risk. Its Lakatosian assessment as progressive remains provisional pending empirical investigation.

> ***The Akimov Theory of Unity is best understood not as a completed theory of consciousness but as a research framework: a set of precisely formulated questions, a rigorous methodology for pursuing them, and a growing body of theoretical and empirical results that progressively narrow the space of viable answers. The investigation of the dynamical conditions of perspectival organization has only begun.***

The second phase of the programme — empirical testing, external peer review, and progressive refinement — now begins. The value of the programme lies not in its claim to completeness, but in the precision with which it defines its scope, risks, and conditions of failure.

The strength of a research programme is not measured by its resistance to criticism, but by its ability to define the conditions under which criticism becomes decisive.

Appendix A: Experimental Protocol for Empirical Validation

This appendix outlines proposed experimental designs intended to test the core predictions of the Akimov Theory of Unity (ToU), specifically the emergence of the perspective attractor through the dynamic coupling of the world-model (M_w) and the self-model (M_s). The protocols are proposed as conceptual frameworks for empirical investigation and may be further developed into detailed experimental specifications.

A.1 Human Neurophysiological Protocol (EEG / TMS / Anaesthesia)

> *Objective: To test empirically whether the coupling between M_w and M_s corresponds to measurable neural integration and phase transitions associated with perspectival organization.*

Participants are subjected to controlled perturbations using transcranial magnetic stimulation (TMS) combined with high-density EEG recording. The stimulation targets regions associated with the two theoretical constructs:

- M_w (world-model): lateral parietal cortex, temporo-parietal junction (TPJ)
- M_s (self-model): medial prefrontal cortex (mPFC), posterior cingulate cortex (PCC)

Three experimental conditions are applied:

- Condition 1 — Baseline wakefulness: full perspectival organization expected.
- Condition 2 — Mild sedation (e.g., low-dose propofol): approaching the critical (M_w, M_s) coupling threshold; critical slowing down and increased variance predicted.
- Condition 3 — Deep sedation (controlled clinical threshold): sub-threshold coupling; bifurcation and collapse of the perspective attractor predicted.

The four empirical predictions of the ToU framework (see Section 4) map onto this paradigm as follows. Critical slowing down should manifest as increasing TMS-evoked EEG response duration approaching the sedation threshold. Increased variance should appear in EEG dynamics in low-frequency bands linked to M_w–M_s coupling circuits. Bifurcation should produce a qualitative discontinuity in the Perturbation Complexity Index (PCI) at the loss-of-consciousness threshold. Hysteresis should produce an asymmetry between the propofol concentration at loss of consciousness and the concentration at recovery.

> *The anatomical specificity prediction constitutes the discriminating test: signature indicators should be concentrated in circuits linking mPFC/PCC with TPJ/lateral parietal cortex, not distributed uniformly across cortex. This anatomical distribution distinguishes ToU predictions from global complexity accounts.*

Measurement instruments

- Functional connectivity: phase synchrony and coherence between M_w and M_s circuits.
- Perturbation Complexity Index (PCI): Lempel-Ziv complexity of compressed TMS-EEG response.
- TMS-evoked potentials: amplitude, duration, and propagation dynamics of cortical responses.
- Recovery time τ_relax: monotonically tracked across graded sedation levels as the primary critical slowing down measure.

A.2 Artificial System Simulation Protocol

Objective: To test whether perspective-like attractor dynamics can emerge in artificial systems satisfying the minimal coupling conditions between world-model and self-model modules as specified by the ToU mechanistic account (Section 3).

A minimal dual-module architecture is constructed in accordance with the theoretical requirements of Section 3:

- M_w module: external data processing module responsible for environmental modelling, producing a dynamically updated representation of the system's environment.
- M_s module: internal state model implementing recursive self-representation — a dynamically updated representation of the system's own state, capacities, and orientational position.

The two modules are bidirectionally coupled with a controllable coupling parameter θ. The system may be implemented using neural network architectures or modular cognitive agents. The coupling strength is varied systematically across the following experimental conditions:

- Sub-threshold ($\theta < \theta_c$): The ToU framework predicts fragmented, unstable representations without attractor convergence. No stable (M_w, M_s) fixed point should form.
- Near-threshold ($\theta \approx \theta_c$): Critical slowing down predicted — increasing recovery time following perturbation; increased variance in system state trajectories.
- Super-threshold ($\theta > \theta_c$): Stable attractor dynamics predicted — robust, self-stabilizing convergence of the (M_w, M_s) joint dynamics to a fixed point A*.

Noise and perturbations are introduced at each coupling level to test attractor stability and relaxation time τ_{relax} as a function of proximity to the critical threshold.

Measurement instruments

- Representation stability: variance of M_w and M_s state vectors over time at each coupling level.
- Information integration metrics: mutual information and transfer entropy between M_w and M_s modules.
- Recurrence analysis: recurrence plots and recurrence quantification of the system state space to detect attractor formation.
- Relaxation time τ_{relax}: time required for the joint (M_w, M_s) state to return to its pre-perturbation trajectory following a standardized input perturbation.

A.3 Cross-Domain Prediction: Substrate Independence

A central theoretical commitment of the ToU framework is substrate independence: the four empirical signatures of the perspective attractor transition are predicted by the mathematical structure of the (M_w, M_s) phase transition rather than by any specific biological or computational implementation.

This commitment generates a cross-domain prediction: biological systems (Protocol A.1) and artificial systems (Protocol A.2) should exhibit structurally equivalent signatures at their respective perspective transitions, provided the minimal dynamical conditions are satisfied. Specifically:

- Critical coupling thresholds should produce analogous bifurcation signatures in both biological and artificial systems.
- The structured co-occurrence of all four signatures — rather than their isolated presence — should be the empirical marker of a genuine (M_w, M_s) attractor transition in both domains.
- Hysteresis should be present in both domains: the threshold to enter the perspectival regime should exceed the threshold to remain in it, regardless of implementation substrate.

> Substrate independence does not imply implementation equivalence. Biological and artificial systems may instantiate the (M_w, M_s) coupling through entirely different physical mechanisms. The cross-domain prediction concerns only the mathematical structure of the transition signatures, not the specific processes that produce them. Any account that explains the signatures in one domain but not the other without theoretical justification would constitute evidence against the substrate-independence claim.

The experimental programme of this appendix is intended as a conceptual foundation for empirical investigation. Full operationalization of the protocols — including sample sizes, stimulus parameters, connectivity analysis pipelines, and statistical thresholds — requires further development in collaboration with empirical researchers in neuroimaging, anaesthesiology, and computational cognitive science.

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